Gummadidala Geetha Charani

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.63
Sri Chaitanya Junior Kalasala	Class XII (Telangana Board of Intermediate Education)	2024	97.00%
Sri Chaitanya School	Class X (Telangana Board of Secondary Education)	2022	9.70%

PROJECTS

AI Agent for Personalized Vital-Sign Interpretation | *LLMs, Agentic AI, RAG, LangChain, Python*

- Developing a **multi-agent AI system for personalized vital-sign assessment**, combining LLM reasoning, clinical rules, and longitudinal patient history.
- Implementing **patient-specific 7-day baselines** to detect deviations from individual vital trends.
- Integrating **RAG-based medical knowledge and patient-history retrieval** with specialized agents for normal, abnormal, emergency, and follow-up assessment.
- Building **clinical copilot and human-escalation workflows** to provide context-aware summaries, targeted follow-up questions, and escalation of severe cases.

Waste Object Detection with YOLOv8 |Python, PyTorch, Ultralytics YOLO

- Developed an object detection pipeline for waste localization and classification using the **TACO** dataset with YOLOv8.
- Converted COCO-format annotations to YOLO format for efficient model training,and consolidated 60 waste categories into **8 super-categories** to address severe class imbalance.
- Trained and evaluated multiple YOLOv8s configurations by varying **image resolution, augmentation techniques**.
- Achieved a mAP@50 of 0.265 on a dataset of ~1,500 images despite significant data scarcity challenges.
- Analyzed detection performance across classes, identifying **class imbalance and limited data** as key limitations.

Binary-Class Image Classification using CNNs and SE Attention (TEAM PROJECT)

- Architected a custom CNN in **PyTorch** with **Residual Blocks** and **SE attention** to classify 3D geometric shapes.
- Optimized training on **18,000 images** using **AdamW**, **OneCycleLR**, and **dropout** to improve convergence and generalization.
- Analyzed and addressed complex test-set distribution shifts caused by surface reflectance and specular highlights.
- Achieved **Rank 20/74** on the competition leaderboard while training entirely **without pre-trained weights**.

Neural Network Architectures and Sequence Modeling | NumPy, PyTorch

- Implemented **CNNs, RNNs, LSTMs, Transformer Encoders, and State Space Models** from scratch using NumPy and PyTorch.
- Developed **BPTT, Multi-Head Self-Attention**, and custom training pipelines for sequence learning tasks.
- Evaluated optimization techniques including **SGD, Momentum, RMSProp, and Adam** across image classification and sentiment analysis tasks.
- Investigated regularization methods such as Dropout and Batch Norm, analyzing their impact on convergence.
- Performed comparative analysis of CNNs, LSTMs, Transformers, and SSMs on image classification, sentiment analysis, and long-context modeling tasks.

RELEVANT COURSES

Artificial Intelligence: Intro to Modern AI , Classical AI , programming for AI , Foundations of Machine Learning, Optimization-1 , Optimization-2 ,Deep Learning, AI for Humanity

Mathematics: Discrete mathematics, Calculus-1 , Calculus-2 , Vector Calculus, Series of Functions, Probability and random variables, Differential equations, Matrix theory, Intro to Metric spaces

Computer Science: Introduction to programming, DBMS 1, DSA , Algorithms

Other: Linear Systems and Signal Processing, Control Systems

SKILLS

Programming Languages: C, Python, MySQL

Python Frameworks & Libraries: Pytorch, Scikit-learn, Numpy, Pandas, Matplotlib

Developer Tools & Operation Systems: Latex, VS Code, GitHub, Git, Windows, Linux

EXTRACURRICULAR

- Silver Medalist - Cricket, Diesta 2026, IIT Hyderabad
- Bonze Medalist - Carrom, Milan 2024, IIT Hyderabad
- Active Badminton Player - National Sports Organisation (NSO), IIT Hyderabad

POSITIONS OF RESPONSIBILITY

Sunshine Buddy 2025-2026 | IITH Mental Health Club

- Supported student well-being initiatives by serving as a peer buddy, fostering engagement and providing guidance to assigned students.

Events Head -- Milan, IIT Hyderabad | 2026

- Led event planning and execution, coordinating **multiple teams, coordinators, schedules, and event requirements**.
- Managed **event guidelines and operations**, ensuring smooth coordination and timely execution of activities.